

CASTELO BRANCO, Portugal

WMO: 085700

Lat: 39.8394N

Lon: 7.4786W

Elev: 1269

StdP: 14.03

Time Zone: 0.00 (EUW)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.9	36.2	20.0	15.7	42.7	23.9	18.9	44.9	20.7	50.7	18.4	49.8	5.6	30	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.6	97.8	66.7	95.0	66.0	92.2	65.3	69.2	89.7	68.1	88.2	67.1	86.3	6.0	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	93.7	70.5	62.7	89.3	70.0	61.4	85.2	69.8	33.9	89.6	33.0	88.4	32.2	86.4	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	15.0	13.2	DB	29.7	102.6	2.4	2.8	27.9	104.6	26.5	106.2	25.1	107.8	23.3	109.8	(4)
(5)			WB	26.8	71.6	2.7	1.0	24.9	72.3	23.3	72.9	21.8	73.5	19.9	74.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	47.3	49.5	54.1	57.1	63.5	71.6	76.4	77.2	71.4	62.8	53.0	48.1	(6)	
	DBStd	12.11	4.65	4.67	5.61	6.29	7.05	7.44	6.38	6.05	6.10	5.87	5.03	4.41	(7)	
	HDD50	315	107	60	23	8	0	0	0	0	0	1	26	90	(8)	
	HDD65	2720	548	433	339	250	117	21	1	12	112	361	525		(9)	
	CDD50	4353	24	47	149	220	417	647	819	844	642	399	115	30	(10)	
	CDD65	1285	0	0	1	12	69	218	356	380	204	45	0	0	(11)	
	CDH74	14324	0	0	13	116	720	2518	4274	4447	1919	317	0	0	(12)	
	CDH80	6787	0	0	0	10	216	1175	2228	2314	763	81	0	0	(13)	
(14) Wind	WSAvg	6.0	5.8	6.4	6.7	6.6	6.1	6.2	6.4	6.1	5.4	5.3	6.0	5.6	(14)	
(15) Precipitation	PrecAvg	26.1	3.2	2.4	2.2	2.3	2.0	0.6	0.2	0.3	1.6	4.3	3.7	3.5	(15)	
	PrecMax	41.6	15.2	6.6	8.9	5.9	6.2	2.0	1.2	1.3	5.0	8.4	11.7	12.1	(16)	
	PrecMin	15.9	0.0	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.1	0.7	0.0	0.4	(17)	
	PrecStd	7.4	3.0	1.7	2.2	1.5	1.8	0.6	0.3	0.3	1.3	2.3	3.0	3.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.5	67.3	75.7	81.4	90.6	99.0	101.1	101.5	95.6	87.7	70.5	62.3	(19)	
		MCWB	52.5	53.1	56.2	61.7	64.6	67.5	67.5	67.4	65.8	61.0	58.4	54.1	(20)	
	2%	DB	59.6	64.1	71.8	77.4	85.7	94.6	97.6	97.9	91.6	81.7	66.5	59.5	(21)	
		MCWB	51.6	52.6	54.6	59.4	63.3	65.9	66.5	66.6	64.4	60.4	57.1	54.1	(22)	
	5%	DB	56.9	60.9	68.6	73.8	81.5	91.2	94.8	94.9	88.0	77.1	63.5	57.4	(23)	
		MCWB	51.4	51.4	53.9	58.3	61.6	65.2	66.3	65.6	63.4	59.9	55.5	53.0	(24)	
	10%	DB	54.8	58.0	65.0	69.5	77.6	87.2	91.6	91.7	84.4	73.0	60.7	55.4	(25)	
		MCWB	50.6	50.5	53.0	56.4	60.5	64.6	65.5	65.1	62.4	59.0	54.9	51.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.1	57.5	59.8	64.2	67.8	70.0	70.8	70.6	68.2	66.9	62.8	58.3	(27)	
		MCDB	58.8	61.5	69.1	78.0	86.0	90.0	91.9	91.8	86.9	74.1	65.9	59.3	(28)	
	2%	WB	54.9	55.4	57.4	61.4	65.3	68.3	69.1	69.1	66.7	65.0	59.8	56.0	(29)	
		MCDB	56.8	59.1	65.3	74.2	80.4	87.9	90.7	90.0	83.5	71.4	62.6	57.7	(30)	
	5%	WB	53.3	53.6	55.8	59.3	63.5	67.0	67.9	67.9	65.6	63.5	58.2	54.2	(31)	
		MCDB	55.3	58.2	63.6	70.9	77.4	86.0	89.3	87.9	81.1	70.0	60.9	56.4	(32)	
	10%	WB	51.5	51.9	54.3	57.4	61.8	65.6	66.7	66.7	64.5	61.9	56.5	52.4	(33)	
		MCDB	53.8	56.7	62.1	67.0	74.5	83.7	87.0	86.4	78.7	68.4	59.4	54.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.1	15.3	17.0	17.7	20.1	23.4	25.8	25.6	21.4	16.5	13.3	12.3	(35)	
		MCDBR	15.9	19.2	22.1	23.3	25.0	26.9	27.6	27.3	25.3	22.7	16.1	14.5	(36)	
	5% WB	MCWBR	10.3	11.0	11.3	10.9	10.5	10.1	9.5	9.5	9.3	9.3	9.1	9.6	(37)	
		MCWBR	11.6	15.1	18.2	20.9	22.4	25.4	26.7	25.7	21.6	15.7	12.0	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.305	0.320	0.351	0.352	0.360	0.362	0.348	0.363	0.353	0.343	0.316	0.304	(40)	
		taud	2.523	2.480	2.405	2.411	2.414	2.423	2.432	2.391	2.447	2.502	2.526	2.528	(41)	
	Ebn at Noon		276	286	287	292	291	289	292	285	282	274	268	267	(42)	
		Edh at Noon	24	28	34	36	37	36	36	36	32	28	23	22	(43)	
(44) All-Sky Solar Radiation	RadAvg		678	986	1326	1675	2065	2366	2479	2182	1662	1091	740	601	(44)	
	RadStd		100	136	180	187	206	134	70	89	120	119	116	87	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page